Why do PCA and classical MDS produce the same embedding?

PCA and Classical MDS start from fundamentally different inputs. PCA operates directly on data coordinates, whereas Classical MDS requires only pairwise distances. Although some applications appear to begin with coordinates, these coordinates are used solely to construct the pairwise distance matrix.

Surprisingly, despite these fundamentally different starting points, the two methods often produce almost identical embeddings for Euclidean datasets. Is this merely a coincidence, or does it reflect a deeper mathematical connection?

Theoretical Relationship Between PCA and Classical MDS

A Step-by-Step Comparison

As a brief review, PCA starts from a centered data matrix

$$X = [x_1, x_2, \ldots, x_m].$$

The basic procedure is

$$X \to XX^T \to XX^T W = W\Lambda,$$

where the columns of $W = [w_1, w_2, \ldots]$are the principal directions. Multiplying $XX^T$by the constant factor $1/(m-1)$simply converts it into the covariance matrix and does not change the eigenvectors.

Classical MDS follows a different route. Starting from the pairwise distance matrix D, it first reconstructs the centered Gram matrix Busing

$$b_{ij} = -\frac{1}{2}\left(d_{ij}^2 - d_{i:}^2 - d_{:j}^2 + d_{::}^2\right).$$

After eigen-decomposition,

$$B = V\Lambda V^T,$$

the reconstructed coordinates are obtained as

$$Z = \Lambda^{1/2} V^T,$$

where the number of retained dimensions is chosen according to the desired embedding dimension.

Therefore, the overall procedure of Classical MDS is

$$D \to B \to V\Lambda V^T \to Z.$$

The MDS Solution Is Not Unique

An important observation is that the coordinate matrix reconstructed by Classical MDS is **not** the unique solution satisfying

$$B = Z^T Z.$$

Given the same distance matrix D, there are infinitely many coordinate systems that represent exactly the same point configuration. These coordinate systems may differ by orthogonal transformations, embedding dimensions, or other equivalent representations, while preserving the same pairwise distances.

Classical MDS constructs one particular representation through the following steps.

First, transforming

$$D \to B$$

implicitly assumes that the reconstructed coordinates are centered. In other words, the origin is fixed at the centroid of the point configuration.

Second, after obtaining the centered Gram matrix, there are still infinitely many coordinate matrices satisfying

$$B = Z^T Z,$$

possibly with different embedding dimensions. Classical MDS selects the embedding dimension specified by the user.

Finally, among all equivalent coordinate systems, Classical MDS adopts the one generated directly from the eigen-decomposition,

$$Z = \Lambda^{1/2} V^T.$$

Therefore, Classical MDS should not simply be regarded as a method for recovering a coordinate representation of the point configuration from pairwise distances. Instead, it selects a canonical coordinate system determined by the eigen-decomposition from an infinite family of equivalent embeddings.

Why Do PCA and Classical MDS Produce the Same Embedding?

The connection between PCA and Classical MDS now becomes clear.

Suppose we perform PCA on the coordinates reconstructed by Classical MDS. Since $Z$is already centered,

$$\begin{aligned} \mathrm{Cov}(Z) &= \frac{ZZ^T}{m-1} \\ &= \frac{\Lambda^{1/2} V^T V \Lambda^{1/2}}{m-1} \\ &= \frac{\Lambda}{m-1}. \end{aligned}$$

which is already diagonal.

Therefore, no further rotation is required—the coordinates reconstructed by Classical MDS are already expressed in the principal-coordinate system.

Now consider another valid reconstruction,

$$Z' = Q \Lambda^{1/2} V^T,$$

where $Q$is an arbitrary orthogonal matrix.

This alternative reconstruction produces exactly the same pairwise distance matrix because

$$\frac{Z' Z'^T}{m-1} = \frac{Q \Lambda Q^T}{m-1},$$

which differs only by an orthogonal transformation.

This explains why the distance matrix admits infinitely many coordinate representations of the point configuration, while Classical MDS chooses the canonical one obtained directly from eigen-decomposition. That canonical representation coincides with the coordinate system produced by PCA.

Underlying Principles

The mathematical derivation explains **how** Classical MDS and PCA produce the same embedding. A more fundamental question is **why** this happens. Why does the canonical embedding selected by Classical MDS coincide with the PCA coordinate system?

The objective of PCA is well known: among all $d' < d$-dimensional linear subspaces, it seeks the one that maximizes the total variance of the projected data.

What, then, is the corresponding objective of Classical MDS?

Starting from

$$B = Z^T Z,$$

consider the quadratic form

$$c^T Z^T Z c = \| Zc \|^2,$$

where $c$satisfies

$$c^T c = 1.$$

The vector Zcrepresents a direction in the reconstructed coordinate system, and its squared Euclidean norm measures the energy along that direction.

Consequently, the solution adopted by Classical MDS is exactly the optimizer of

$$\operatorname{argmax} c^T Z^T Z c,$$

subject to

$$c^T c = 1.$$

More generally, if we seek a d'-dimensional subspace, the optimization becomes

$$\operatorname{argmax} \operatorname{Tr} (C^T Z^T Z C),$$

subject to

$$C^T C = I.$$

Applying the method of Lagrange multipliers yields

$$Z^T Z C = C\Lambda.$$

Therefore, Classical MDS can be interpreted as searching for the set of directions that maximizes the total energy of the point configuration under the Euclidean-distance constraint.

This immediately explains the equivalence with PCA. Formally, the matrices ZZ^T and $Z^T Z$ generally have different dimensions and different eigenvectors, but they share exactly the same nonzero eigenvalues, and their eigenvectors are related through the singular vectors of Z. Conceptually, PCA and Classical MDS solve the same optimization problem from different mathematical formulations. PCA maximizes the covariance of the projected data, whereas Classical MDS maximizes the energy of the reconstructed point configuration. For centered Euclidean data, these two objectives are equivalent. Therefore, both methods ultimately solve the same spectral problem and recover equivalent embeddings.

Experimental Results on a Toy Dataset

To verify the theoretical analysis, we first construct a simple two-dimensional toy dataset. A total of 1,000 random points are generated within the unit square, with the additional constraint that each point lies within a distance of 0.2 from the line $y = 0.5$. Consequently, the data approximately lie on a one-dimensional manifold embedded in a two-dimensional space. The original dataset is shown in Figure 1.

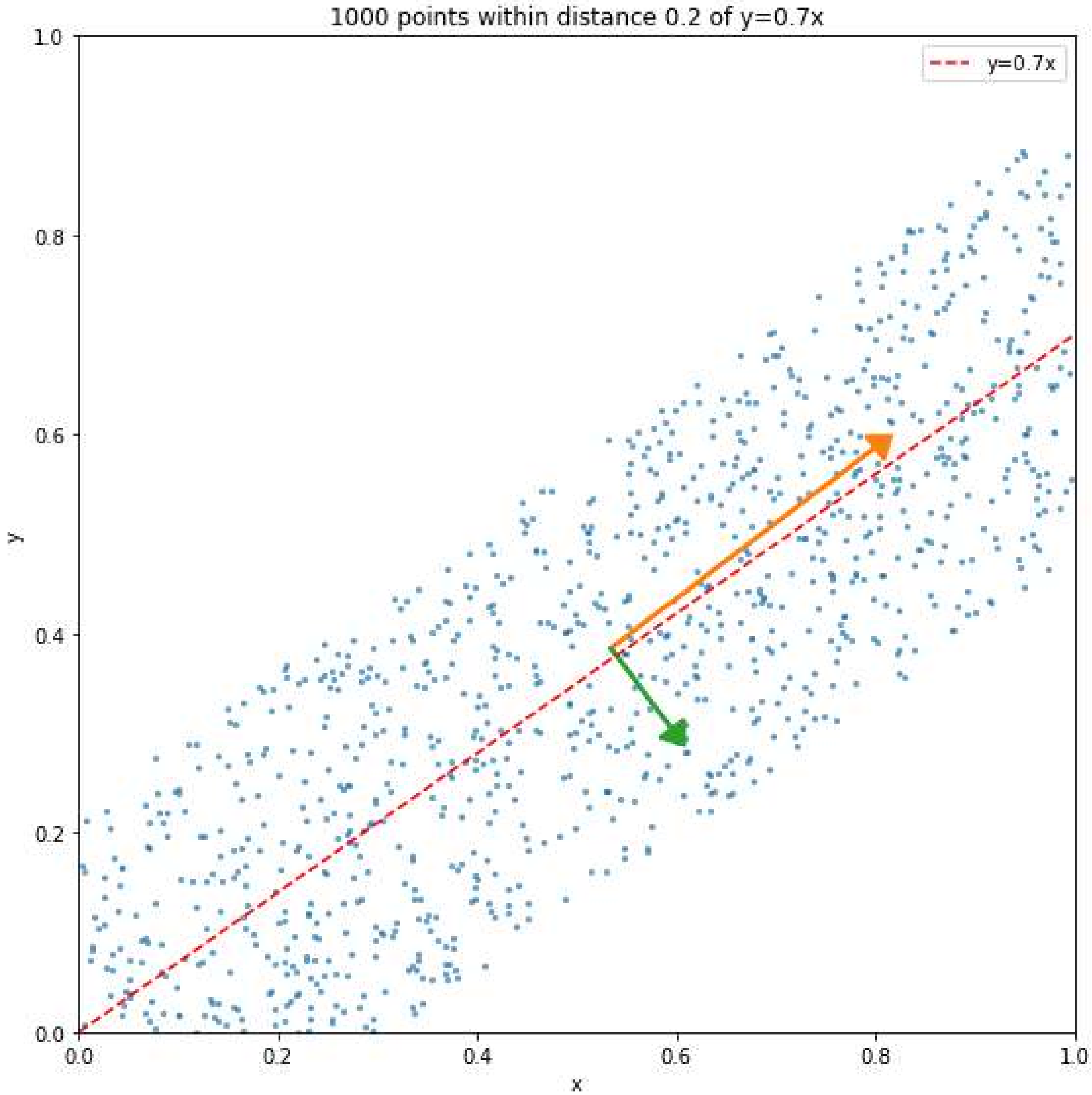

Fig. 1 The generated toy data points

To make the comparison more intuitive, we first perform PCA on the original coordinates. The resulting principal components are illustrated in Figure 1. The corresponding eigenvalues are

$$\lambda_1 = 110.7,\ \lambda_2 = 10.26.$$

These values will be compared with those obtained from Classical MDS later.

Next, we compute the pairwise Euclidean distance matrix from the same dataset and apply Classical MDS. The reconstructed embedding is shown in Figure 2.

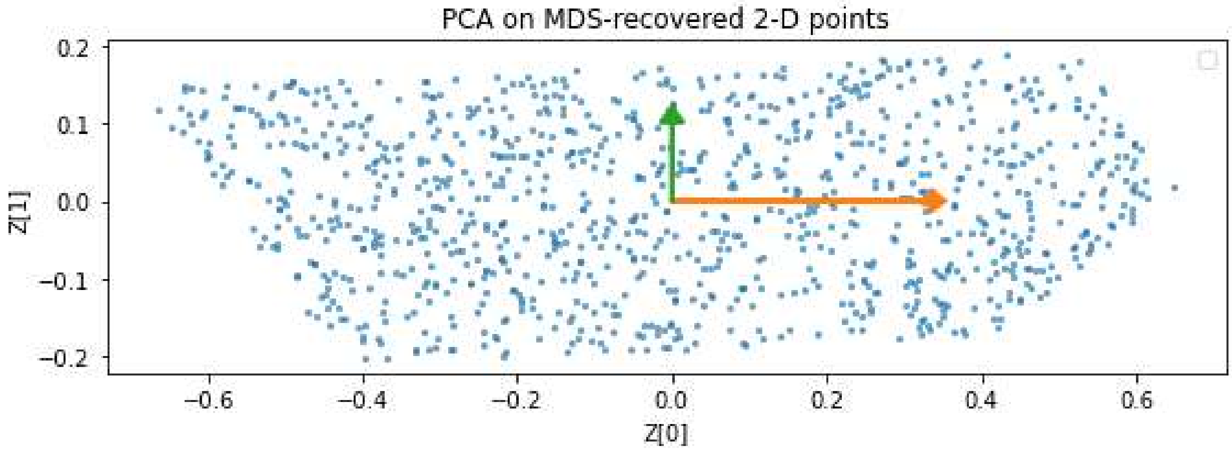

Fig. 2 The reconstructed embedding from MDS

Visually, the embeddings produced by PCA and Classical MDS are almost indistinguishable. As expected from the theoretical analysis, the two methods recover the same principal-coordinate representation, differing only by possible sign changes or other orthogonal transformations, which do not alter the underlying geometry of the point configuration.

To further verify the equivalence, we compare the eigenvalues obtained from both methods. The eigenvalues recovered by Classical MDS are

$$\lambda_1 = 110.7,\ \lambda_2 = 10.26,$$

which are identical to those obtained from PCA, up to numerical precision. This agrees exactly with the theoretical result that the nonzero eigenvalues of ZZ^T and $Z^T Z$ are the same.

Finally, we compare the principal directions recovered by the two methods. As predicted, the directions coincide up to an orthogonal transformation (or, in this two-dimensional example, a possible sign flip). This confirms that PCA and Classical MDS recover the same principal-coordinate system from two different mathematical descriptions of the same Euclidean point configuration.

Experimental Results on the Yale Face Dataset

To further validate the theoretical analysis, we next consider a real-world dataset. The Yale Face Database is a standard benchmark in pattern recognition and dimensionality reduction. Unlike the toy example, this dataset consists of high-dimensional face images with variations in facial expression and illumination, providing a more realistic evaluation of the proposed analysis.

Fig. 3 Examples of Yale Face database

Each image is first reshaped into a vector, and the resulting data matrix is centered before applying PCA. The first two principal components are used for visualization, and the resulting embedding is shown in Figure 4.

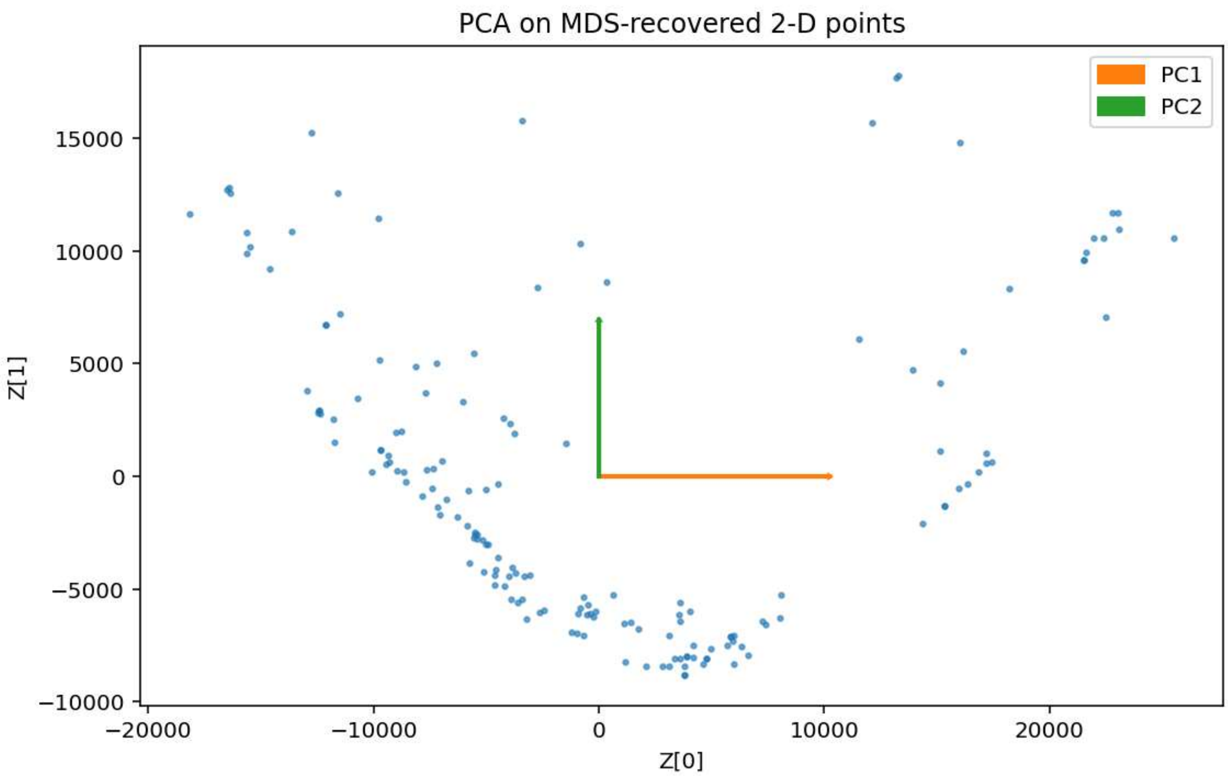

Fig. 4 The embedding's first two dimensions of Yale faces database.

Using the same dataset, we then compute the pairwise Euclidean distance matrix and apply Classical MDS with the same embedding dimension, and we can obtain exactly the same embedding as in Figure 4.

Despite starting from entirely different mathematical descriptions—coordinates for PCA and pairwise distances for Classical MDS—the two embeddings exhibit almost identical geometric structures. Images corresponding to similar facial expressions and lighting conditions remain close to one another in both embeddings, while images with larger appearance differences are consistently placed farther apart.

To further quantify the agreement, we compare the eigenvalues obtained from the two methods. The first three nonzero eigenvalues recovered by Classical MDS match those obtained from PCA to numerical precision, providing additional evidence that both methods solve the same underlying spectral problem.

$$\lambda_1 = 1.71\text{E}10, \lambda_2 = 7.90\text{E}9, \lambda_3 = 4.99\text{E}9$$

Overall, the Yale Face experiment demonstrates that the theoretical relationship derived in the previous section is not restricted to simple synthetic data. It remains valid for realistic high-dimensional datasets, where PCA and Classical MDS continue to recover equivalent principal-coordinate representations from two different mathematical descriptions of the same Euclidean point configuration.

Discussion

The theoretical derivation and experimental results establish that PCA and Classical MDS produce equivalent embeddings for centered Euclidean data. However, the significance of this equivalence extends beyond the mathematical proof itself. More importantly, it provides a unified perspective on how these two seemingly different methods should be understood.

PCA and Classical MDS are often introduced as two independent dimensionality reduction algorithms. PCA starts from data coordinates, whereas Classical MDS starts from pairwise distances. From an algorithmic point of view, they appear to solve different problems. A more unified view, however, is to regard them as two computational routes to the same centered Euclidean point configuration.

The key observation is that coordinates and pairwise distances are not competing descriptions of the data. Instead, they are two different mathematical descriptions of the same underlying Euclidean point configuration. Coordinates explicitly specify the location of each point in a coordinate system, whereas pairwise distances describe only the geometric relationships among the points. Although these descriptions contain different forms of information, they characterize the same underlying geometry.

The centered Gram matrix serves as the bridge between these two descriptions. PCA reaches the Gram matrix directly from the coordinates, while Classical MDS reconstructs it from pairwise distances through double centering. Once the centered Euclidean point configuration has been encoded in the Gram matrix, both methods reduce to the same eigen-decomposition problem and therefore recover the same principal-coordinate representation. We can use the following figure to describe the concept relations.

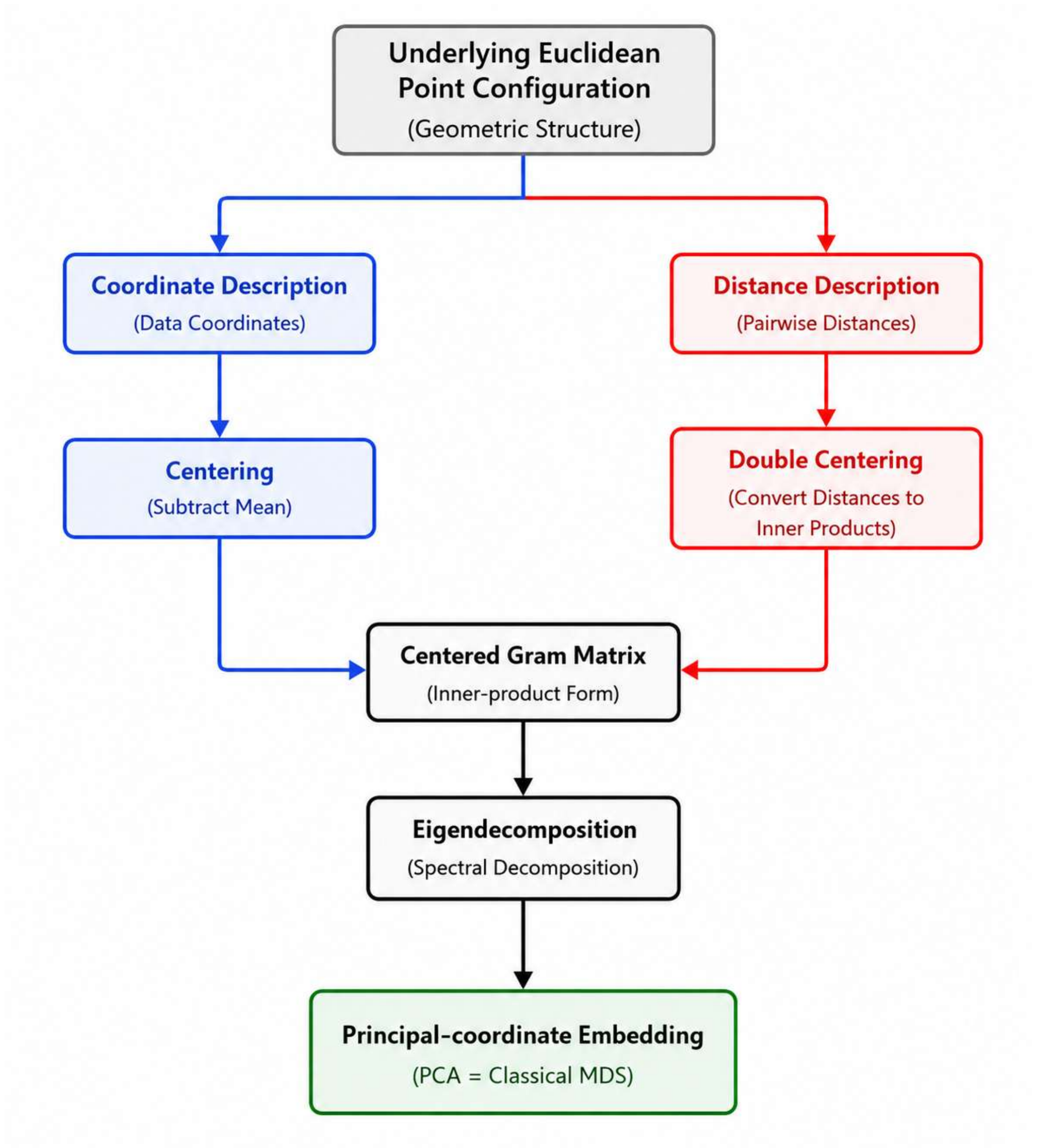

Fig. 5 Coordinates and pairwise distances provide two different mathematical descriptions of the same Euclidean point configuration. PCA and Classical MDS take these two descriptions as input, but both ultimately recover the same principal-coordinate representation through the centered Gram matrix.

This unified viewpoint also clarifies the scope of the equivalence. The discussion throughout this paper applies specifically to **Classical MDS** under **Euclidean distances**. If the dissimilarities are non-Euclidean, the double-centered matrix is no longer guaranteed to represent an exact

Euclidean point configuration. Likewise, metric and non-metric MDS optimize different objective functions and therefore are not expected to be equivalent to PCA in general.

More broadly, this perspective suggests that many dimensionality reduction algorithms differ primarily in **how the geometric structure of the data is represented**, rather than in **what geometric structure they ultimately seek to recover**. In the case of PCA and Classical MDS, the mathematical descriptions are different, but the underlying geometry—and consequently the recovered principal-coordinate representation—is the same.

Summary

PCA and Classical MDS appear to start from different mathematical descriptions, yet they ultimately recover equivalent principal-coordinate embeddings for centered Euclidean data.

This equivalence is not an algebraic coincidence. It reflects the fact that coordinates and pairwise distances describe the same underlying Euclidean point configuration from different perspectives.

Different mathematical descriptions, the same underlying geometry.